# MOHAMED AMINE KOUBAA

## Full Stack & DevOps Engineer - AWS Cloud, CI/CD & AI Solutions

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## PROFILE

Full Stack and DevOps engineer with 5 years of experience, mostly on AWS. I take a product from start to finish: API design, Angular front end, Terraform infrastructure, GitLab pipeline and follow-up once it is live. I also work on generative AI projects (enterprise LLM gateway, agents, MCP servers). Quality matters to me: test-driven development (TDD) and automated tests with coverage checks in every pipeline.

## KEY SKILLS

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|---|---|
| **Cloud & IaC** | AWS (Lambda, API Gateway, S3, SQS, Cognito, EventBridge, KMS, CloudFront, Rekognition, SES, IAM, CloudWatch, EC2, CloudFormation, Elastic Beanstalk) · Terraform · Terragrunt |
| **DevOps & CI/CD** | GitLab CI · Git · ArgoCD (GitOps) · HashiCorp Vault |
| **Containers & Monitoring** | Docker · Kubernetes · NGINX Ingress · CloudWatch · Langfuse (LLM tracing) · Structured logging |
| **Backend** | Python (Chalice, FastAPI) · Node.js · NestJS · Express · REST APIs · Microservices · Event-driven architectures (SQS) · Pydantic |
| **AI & Data** | Generative AI · LLM · AI agents (LangGraph) · Model Context Protocol (MCP) · JSON-RPC · Embeddings and vector search · Apache Airflow · Amazon Rekognition |
| **Frontend** | Angular · React.js · Next.js · TypeScript · JavaScript · HTML · SCSS |
| **Databases** | MongoDB (Atlas Vector Search) · MySQL / MariaDB · SQL · NoSQL · Sequelize |
| **Mobile** | Flutter · Dart |
| **Testing & Methods** | TDD · Automated unit and integration tests in CI/CD pipelines with coverage gates · pytest · Vitest · Karma / Jasmine · Playwright · Agile / Scrum |

## CERTIFICATIONS

- AWS Certified Developer – Associate - AWS, 2026
- Docker Certified Associate (DCA) - Docker, 2026
- Astronomer Certified for AI Orchestration Fundamentals - Astronomer, 2026
- Model Context Protocol: Advanced Topics - Anthropic, 2026
- Introduction to Model Context Protocol - Anthropic, 2026
- Multicloud Network Associate - Aviatrix, 2025

## EDUCATION

- Engineering degree in Software Engineering - ESPRIT, Tunisia, 2021-2024
- Applied Bachelor's degree in Computer Science - ISTIC Borj Cedria, Tunisia, 2018-2021

## LANGUAGES

- French: TCF - C1
- English: TOEFL - B2

## PROFESSIONAL EXPERIENCE

### Full Stack & DevOps Engineer, Portago

*12/2024 - Present | France*

**LLM Gateway - Enterprise gateway between products and AI providers**

- Built the backend of an OpenAI-compatible gateway that routes every AI call in the company: 16 Lambda services in Python/Chalice, per-application routing, automatic failover to another provider during outages and streamed responses (SSE).
- Put cost and security controls in place: per-application budgets with the cost reserved before each call, access policies, Langfuse tracing and a usage ledger for FinOps reporting.
- Connected MCP servers with tools ranked by risk level, and shipped the Angular admin console (FR/EN/AR).

- Managed the infrastructure across 3 environments (Terraform/Terragrunt), the GitLab pipelines, Vault secrets and KMS encryption of provider keys.

*Technologies: Python · Chalice · AWS Lambda · API Gateway · Cognito · SQS · KMS · MongoDB · Angular · Terraform · GitLab CI · Vault · Langfuse · MCP*

**Moveo - Talent, skills and internal mobility platform**

- Designed the serverless services (Lambda/Chalice, SQS, Cognito) and the asynchronous jobs that use AI to generate learning paths, quizzes, career recommendations and CV and job offer analysis.
- Built the platform assistant: a LangGraph agent and a 132-tool MCP server, deployed on Kubernetes with ArgoCD and connected to the models through the LLM gateway.
- Worked on the Angular front end (employee, HR and admin areas) and on the engine that matches skills to positions.

*Technologies: Python · Chalice · FastAPI · Generative AI · LangGraph · MCP · Kubernetes · ArgoCD · Angular · MongoDB · SQS · Terraform*

### Full Stack & DevOps Developer, Allence Cloud

*11/2021 - 11/2024 | Tunisia*

**Career - AI-assisted recruitment platform**

- Built the full hiring flow: job offers published from a link, text, image or form, CV analysis by LLM, a score out of 100 and AI-led interviews in writing, by voice and by phone (Twilio, OpenAI Realtime).
- Mapped the skills extracted from CVs to a catalogue of about 530 skills, later shared with Moveo, and added a side-by-side comparison of two candidates for recruiters.
- Chose an event-driven design (SQS queues and Lambda workers) to handle long-running jobs, with FastAPI microservices on Kubernetes and MongoDB Atlas vector search.
- Developed with TDD (pytest, pytest-asyncio) and added a coverage check to the GitLab pipelines that blocks delivery below the agreed threshold.

*Technologies: Python · Chalice · FastAPI · Generative AI · MCP · Angular · MongoDB Atlas · SQS · Kubernetes · ArgoCD · Twilio · Terraform · GitLab CI*

**PhotoEvent - Event photography marketplace**

- Built the serverless backend in Python/Chalice (17 Lambda functions, 104 API routes) and the Angular front end, on MongoDB, S3 and SQS.
- Split the backend by business domain (one service, one repository and one pipeline each) so each part ships without breaking the others; pytest tests with an 80 % coverage gate and Trivy image scans.
- Automated gallery imports with an Apache Airflow DAG on Kubernetes and handled photographer finances (balance, commissions, payouts).
- Let guests find their own photos with face recognition (Amazon Rekognition); protected images with watermarks and short-lived signed links.
- Added AI moderation of reviews and automated deployment (Terraform/Terragrunt, GitLab CI, Vault).

*Technologies: Python · Chalice · Generative AI · Angular · MongoDB · S3 · SQS · Rekognition · Airflow · Kubernetes · Terraform · Terragrunt · GitLab CI*

### Full Stack Developer, Dida Mind

*02/2021 - 07/2021 | Tunisia*

**ProHunting - Hunting tourism platform**

- Built the public site in three languages (FR/EN/DE), the client area, a private forum and the back office with role-based permissions.
- Developed the Node.js/Express backend (MySQL, Sequelize) and the Next.js front end, real-time messaging (Socket.IO), PDF document generation and hotel booking through a partner API.
- Gave the operations team the tools they needed: booking and hunting permit tracking, quotes and payments, dashboards, PDF/Excel exports, and content and SEO management.

*Technologies: Node.js · Express · Next.js · TypeScript · MySQL · Sequelize · Socket.IO · Puppeteer · Tailwind CSS · Mapbox · nginx*